Circuit Simulation Lab:32
Design of Sequence Detector

Design a sequential circuit (sequence detector) has a serial input X and an output Z. When the circuit receives a sequence of four consecutive ones (1111) on input X, it produces an output Z = 1 at the same time as the fourth input. At all other times, the output Z = 0. After the circuit receives a successful input sequence of 1111 (which produces Z = 1), a new input sequence of 1111 must be received again to produce Z = 1. (That is, the successful input sequences of 1111 must be non-overlapping.)

CODE:

```
module seq_detector_1111(
input clk,
input rst,
input x,
output reg z
);

reg [2:0] state;

parameter S0 = 3'b000;
parameter S1 = 3'b001;
parameter S2 = 3'b010;
parameter S3 = 3'b011;
parameter S4 = 3'b100;

always @(posedge clk)
begin

if(rst)
begin
   state <= S0;
   z <= 0;
end

else
begin

case(state)
```

```
S0:
begin
   z <= 0;
   if(x)
      state <= S1;
   else
      state <= S0;
end

S1:
begin
   z <= 0;
   if(x)
      state <= S2;
   else
      state <= S0;
end

S2:
begin
   z <= 0;
   if(x)
      state <= S3;
   else
      state <= S0;
end

S3:
begin
   if(x)
   begin
      state <= S0;
      z <= 1;
   end
   else
   begin
      state <= S0;
      z <= 0;
   end
end

default:
begin
   state <= S0;
```

```
    z <= 0;
end

endcase

end

end

Endmodule
```

TESTBENCH

```
module tb_seq_detector_1111;

reg clk;
reg rst;
reg x;

wire z;

seq_detector_1111 dut(
.clk(clk),
.rst(rst),
.x(x),
.z(z)
);

always #10 clk = ~clk;

initial begin

$dumpfile("seq1111.vcd");
$dumpvars(0,tb_seq_detector_1111);

clk = 0;
rst = 1;
x   = 0;

#20;
rst = 0;

// 001110
x=0; #20;
x=0; #20;
```

```
x=1; #20;
x=1; #20;
x=1; #20;
x=0; #20;

// 1111 detected
x=1; #20;
x=1; #20;
x=1; #20;
x=1; #20;

// 0011
x=0; #20;
x=0; #20;
x=1; #20;
x=1; #20;

// another 1111
x=1; #20;
x=1; #20;
x=1; #20;
x=1; #20;

#40;

$finish;

end

initial
$monitor(
"time=%0t x=%b z=%b",
$time,x,z
);

Endmodule
```

```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_32$ iverilog -o out seq_detector_1111.v tb_seq_detector_1111.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_32$ vvp out
VCD info: dumpfile seq1111.vcd opened for output.
time=0 x=0 z=x
time=10 x=0 z=0
time=60 x=1 z=0
time=120 x=0 z=0
time=140 x=1 z=0
time=210 x=1 z=1
time=220 x=0 z=1
time=230 x=0 z=0
time=260 x=1 z=0
time=330 x=1 z=1
time=350 x=1 z=0
time=410 x=1 z=1
```